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#include <iostream>
using namespace std;
// Class definition for Complex numbers
class Complex
{
private:
float real; // Real part
float imag; // Imaginary part
public:
// Default constructor
Complex()
{
real = 0;
imag = 0;
}
// Parameterized constructor
Complex(float r, float i)
{
real = r;
imag = i;
}
// Operator overloading for addition
Complex operator+(Complex c)
{
Complex temp;
temp.real = real + c.real; // Add real parts
temp.imag = imag + c.imag; // Add imaginary parts
return temp;
}
// Operator overloading for multiplication
Complex operator*(Complex c)
{
Complex temp;
//  $(a+bi)(c+di) = (ac - bd) + (ad + bc)i$ 
temp.real = real * c.real - imag * c.imag;
temp.imag = real * c.imag + imag * c.real;
return temp;
}
// Friend function to overload >> for input
friend istream& operator>>(istream &in, Complex &c)
{
cout << "Enter real part: ";
in >> c.real;
cout << "Enter imaginary part: ";

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in >> c.imag;
return in;
}
// Friend function to overload << for output
friend ostream& operator<<(ostream &out, Complex c)
{
out << c.real;
if (c.imag >= 0)
out << " + " << c.imag << "i";
else
out << " - " << -c.imag << "i";
return out;
}
};
// Main function
int main()
{
Complex c1, c2, sum, product;
cout << "--- Input first complex number ---\n";
cin >> c1;
cout << "\n--- Input second complex number ---\n";
cin >> c2;
// Perform addition
sum = c1 + c2;
// Perform multiplication
product = c1 * c2;
// Display results
cout << "\nSum of complex numbers: " << sum << endl;
cout << "Product of complex numbers: " << product << endl;
return 0;
}

```

OUTPUT

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--- Input first complex number ---
Enter real part: 3
Enter imaginary part: 2
--- Input second complex number ---
Enter real part: 1
Enter imaginary part: 4
Sum of complex numbers: 4 + 6i
Product of complex numbers: -5 + 14i
--- Input first complex number ---
Enter real part: 3
Enter imaginary part: 2
--- Input second complex number ---

```

Enter real part: 1

Enter imaginary part: 4

Sum of complex numbers: $4 + 6i$

Product of complex numbers: $-5 + 14i$